Ravishankar Anirudh

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Research Interests

- Cosmological hydrodynamical simulations of high-redshift galaxies
- Galaxy kinematics and dynamics
- Bursty star formation in early dwarf galaxies
- Comparing simulations to observations via forward modelling of gas emission, photometry and spectroscopy

Education

October 2023 – 2027 (expected)
- **Doctoral student of Astronomy and Cosmic Physics**
  at the Max Planck Institute for Astronomy / IMPRS-HD /Universität Heidelberg, Germany
  Supervisor: Dr. Annalisa Pillepich
  Thesis title (tentative): *Kinematics and galaxy growth in the early Universe: insights from cosmological simulations*

August 2018 – July 2023
- **Bachelor of Science - Master of Science Dual Degree** in Physics (GPA: 8.6)
  at the Indian Institute of Science Education and Research (IISER), Tirupati, India
  Supervisors: Dr Melanie Kaasinen and Dr Gergö Popping (ESO, Garching)
  Thesis title: *Using radiative transfer models and cosmological simulations to map the emission of cold gas within galaxies* (GPA: 9.3).

Research Publications

Journal Articles

1. **R. Anirudh**, A. de Graaff, F. Lacroix, S. H. Price, and A. Pillepich, "Testing the inference of kinematics from mock JWST NIRSpec/MSA observations of TNG50 galaxies at z $\sim$ 2-6," *Monthly Notices of the Royal Astronomical Society*, vol. 550, stag1286, Aug. 2026, ISSN: 0035-8711. DOI: 10.1093/mnras/stag1286
2. **R. Anirudh**, M. Kaasinen, G. Popping, D. Narayanan, K. Garcia, and D. Valentin-Martinez, "Predicting the resolved CO emission of z = 1-3 star-forming galaxies," *Astronomy and Astrophysics*, vol. 699, A19, Jul. 2025, ISSN: 0004-6361. DOI: 10.1051/0004-6361/202453057
3. K. Garcia, D. Narayanan, G. Popping, **R. Anirudh**, S. Sutherland, and M. Kaasinen, "SLICK: Modeling a Universe of Molecular Line Luminosities in Hydrodynamical Simulations," *The Astrophysical Journal*, vol. 974, p. 197, Oct. 2024, ISSN: 0004-637X. DOI: 10.3847/1538-4357/ad642c
4. **R. Anirudh**, C. Eswaraiah, S. Jiao, and J. Jose, "Role of magnetic fields in the fragmentation of the Taurus B213 filament into Sun-type star-forming cores," *Journal of Astrophysics and Astronomy*, vol. 44, p. 59, Dec. 1, 2023, ISSN: 0250-6335. DOI: 10.1007/s12036-023-09948-6

Manuscripts In Preparation

1. **R. Anirudh**, A. Pillepich, D. Nelson, M. C. Smith, and B. Saha, *Discy galaxies with bursty and efficient star formation at $z > 5.5$ in the VESTRAL simulations*, 2026. URL: https://datashare.mpcdf.mpg.de/s/C3cNBHWoap2mKKn
2. D. Nelson, M. C. Smith, A. Pillepich, **R. Anirudh**, and B. Saha, *VESTRAL: Resolved ISM, sub-parsec, single star simulations of high-redshift galaxy formation in the era of JWST*, 2026.
3. A. Pillepich, D. Nelson, M. C. Smith, and **R. Anirudh**, *VESTRAL: Nuclear star clusters at $z > 5.5$ in single-star cosmological zoom-in simulations*, 2026.
4. B. Saha, A. Pillepich, D. Nelson, M. C. Smith, and **R. Anirudh**, *VESTRAL: ISM dynamics driven by resolved supernova explosions in the first billion years*, 2026.

Selected Conferences and Workshops

Contributed Talks

September 2026 • **BUilding Galaxies from Scratch (BUGS) II**, Heidelberg

June – July 2026 • **European Astronomical Society Annual Meeting**, Lausanne *(three talks)*

June 2026 • **Zoom-in views of galaxy disks - Nordita workshop**, Stockholm *(two talks)*

June 2024 • **The feedback-driven matter cycle in galaxies**, Hengstberger Symposium, Heidelberg

Collaboration Meeting Talks

May 2025 • **RUBIES Meeting**, Bergen

July 2025 • **CONDOR Meeting**, Heidelberg

Contributed Posters

September 2026 • **BUilding Galaxies from Scratch (BUGS) II**, Heidelberg

August 2024 • **Observing and Simulating Galaxy Evolution in the Era of JWST**, Ascona

July 2024 • **European Astronomical Society Annual Meeting**, Padova *(two posters)*

February 2024 • **BUilding Galaxies from Scratch (BUGS) I**, Vienna

Observing Proposals

- **JWST Cycle 6**: Co-I on a proposal (submitted) targeting ∼350 galaxies at z ∼4-7 using NIRSpec/MSA slit-stepping

Technical Skills

Codes • AREPO, Python, bash, git, HPC Slurm, LaTeX, C++ (basic), Fortran (basic), Html, css (basic)

Simulations • IllustrisTNG, VESTRAL, Simba

Astro. software • ADQL, astropy, CAESAR, Despotic, DS9, FITS, h5py, msafit, photutils, pysersic, TOPCAT, yt

Instruments • JWST NIRCam and NIRSpec/MSA forward modelling, Gaia, JCMT SCUBA-2, ZTF

Mentoring Experience

Co-supervision

July – August 2026 • **Summer internship** of Michelle Park
Project title: *Interactions drive bursty, spatially clustered star formation in VESTRAL*

February – July 2026 • **Master thesis project** of Jehanne Delhomelle
Thesis title: *The effects of galaxy interactions on star formation at cosmic noon with TNG50*

Tutoring

July – August 2026 • **Laboratory course tutor** at Universität Heidelberg
Supervised the wavefront analysis experiment for 6 groups of undergraduate students

September 2024 and 2025 • **Postgraduate course tutor** at Universität Heidelberg *(twice)*
Supervised the Cosmology block course tutorial sessions for Master students

Academic and Community Service

September 2026 • **Local Organizing Committee Member** of the "BUGS II" conference in Heidelberg

August 2024 – Present • **Working Conditions workgroup Coordinator** of the PhDNet of the Max Planck Gesellschaft, Germany

October 2024 – September 2026 • **External Student Representative** of the Max Planck Institute for Astronomy

October 2024 – September 2025 • **Internal Student Representative** of the Max Planck Institute for Astronomy